

DrivableNets: Drivable Area Detection for Unstructured Environments

Vibhanshu Singh Sindhu
Computer Vision Laboratory
Delhi Technological University
Delhi, India
vibhanshusindhu@gmail.com

Indu Sreedevi
Department of Electronics and
Communication Engineering
Delhi Technological University
Delhi, India
s.indu@dce.ac.in

Ayesha Choudhary
School of Computer & Systems Sciences
Jawaharlal Nehru University
Delhi, India
ayeshac@mail.jnu.ac.in

Abstract—With the advent of autonomous driving over the last decade, many multi-tasking methodologies have been proposed for resolving panoptic perception challenges associated with autonomous driving and have achieved encouraging results as well. Though, most of the work has been done with respect to structured environments, i.e., where the roads are in pristine conditions. These types of networks cannot be efficiently applied to unstructured environments, where the conditions of roads are of subpar level and differ greatly from those of roads in structured environments. Thus, this paper proposes a new and efficient network, capable of performing the multiple tasks of lane detection, traffic object detection and drivable area segmentation simultaneously for unstructured environments, called DrivableNets. Our model accomplished the new state of the art accuracy on the exacting Indian Driving Dataset.

Keywords—Panoptic perception, autonomous driving, multitask learning, drivable area segmentation, traffic object detection, lane line detection.

I. INTRODUCTION

Research in autonomous driving has grown exponentially since the last decade, especially with great development in the field of computer vision and the rise of deep learning. For a vehicle to be proficient in self-driving, it requires all of the information available in its surroundings. Conventionally, techniques such as lidar and radar were employed to gather the information required for the purpose of self-driving. But, because of the high cost associated to these conventional methods and the development of low-cost vision-based tasks which showed that simply using the data from a camera attached to the vehicle can provide sufficient information for a car to be autonomous, there has been a shift to develop complete vision-based approaches for the requirement of autonomous driving. Basically, a panoptic perception system is developed which extracts information from the pictures taken by the camera attached to the vehicle and thus aids the driving system in taking the decisions for the self-driving of the car. Specifically, for a vehicle to be autonomous, it requires the data of all the objects on the road so that it can steer clear of the obstructions and adhere to the traffic rules, and the data of the condition of the road on which it is driving, so as to strategize the driving path of the automobile. Consecutively, these tasks are broadly classified into lane segmentation, drivable area segmentation and traffic object detection.

There have been many great approaches to solve these three tasks individually over the years. For object detection, works like Faster R-CNN [1], SSD [2], CenterNet [3] and YOLO family [4,5,6,7] were proposed. For drivable area segmentation, semantic segmentation models are generally used, such as UNet [8], SegNet [9] and PspNet [10]. While, for lane line segmentation/detection, networks such as LaneNet [11], SCNN [12] and ENet-SAD [13] were proposed. Though these networks have individually achieved great accuracy for their respective tasks, for the task of real-time autonomous driving we need a network that would take into account the limited computational resources available on the devices commonly used in self-driving vehicles and also the latency that occurs due to passing a picture through three networks. Therefore, to overcome these hinderances many multi-tasking models have been proposed by combining the individual models into an encoder-decoder pattern such as DLT-Net [14], MultiNet [15], HybridNets [16], YOLOP [17] and YOLOPv2 [18], where the encoder shares information among different tasks heads to process for specific task.

Though these multi-tasking models have achieved good accuracy, none of them takes into account the unstructured environments of roads in certain areas such as rural areas, where the lane lines are not as smoothly visible, the roads have several potholes, muddy areas, and other obstacles that make the part of the road non-drivable. These complications need special attention, and therefore, a new network is required that can detect the drivable area for an autonomous vehicle in an unstructured environment. Key importance is offered to the dataset employed in the training of this architecture. Since the network aims to detect drivable area for unstructured environments, the dataset of structured environments can not be utilized for the training of our architecture and therefore the Indian Driving Dataset [19] is used which contains of images of roads in unstructured environments, instead of the prominent BDD100K dataset [20] which contains images of roads in structured environments. The intended model is effective for real-time environments such as different lighting conditions, weather variations, and for several kinds of unstructured irregularities found on roads.

The chief inspiration for this work is the pushing need to provide a model that can readily detect the drivable area in an

unstructured environment so as to aid the driving of autonomous vehicles in these areas.

The chief objective of this work is to create a architecture that can accomplish the three tasks: drivable area segmentation, lane segmentation and traffic object detection in unstructured environments, in turn to achieve the ultimate goal of providing a panoptic perception driving system for unstructured environments.

The outline of the rest of the paper is as follows. Section II discusses in detail the previous related works. Section III explains the proposed methodology for our network. Section IV outlines the experimentation design and evaluation. Section V lists the conclusion and future areas of work.

II. LITERATURE ANALYSIS

Researchers have been working in the domain of object detection and semantic segmentation for long and have developed architectures which work very effectively for these individual tasks. We will first review the models proposed to solve the aforementioned three tasks individually and then review the latest multi-tasking learning networks proposed to accomplish these three tasks through a single network.

A. Traffic Object Detection Models

Conventionally, the object detection models typically comprised of three stages. Though today, the existing object detection models can be mostly categorized into one-stage detectors and two-stage or region-based detectors.

The two-stage detectors, as the name suggests, consists of two stages and are relatively more complicated. The first stage consists of screening out all positive examples by performing preliminary tests and generates regions of interest (RoIs). Then, on these RoIs, location refinement and region-based classification is performed in the second stage. Some of the popular two-stage object detectors are Faster R-CNN, FPN [21], R-FCN [22]. In Faster R-CNN, Ren et al. proposed the use of Region Proposal Network (RPN) instead of the Selective Search method used in its predecessor Fast R-CNN, which resulted in increased speed as well as accuracy for object detection. In FPN, Lin et al. proposed the efficient extraction of features of various dimensions from an image by the use of conventional CNN models. In R-FCN, Jifeng et al. significantly improved the speed of detection while maintaining competitive accuracy, by directly using the feature maps to derive RoIs.

While, the one-stage detector algorithms complete the detection in one stage only, by carrying out bounding box regression and classification of objects concurrently. Some of the popular one-stage object detectors are SSD, YOLO family and CenterNet. In YOLO [4], the image is divided into $S \times S$ grids, which considerably accelerates object detection pace and accuracy. In YOLO, the recall of detection is improved by introducing the anchor mechanism. YOLOv3 [5] achieves multi-scale detection by using FPN structure. In YOLOv4 [6], instead of FPN utilized in YOLOv3, a modified PAN and SSP were utilized as the neck, along with data augmentation which made the network faster and more efficient. The latest version of YOLO family, YOLOv7 [7] further enhances the speed and

detection accuracy by providing a more powerful network, a more vigorous loss function and an improved label assignment.

As observed, the one-stage detectors take less computational power, operates faster because it does not have to create regions and thus are favoured in real-time functional uses.

B. Drivable Area Segmentation Models

With the rise in deep learning, several CNN-based semantic segmentation models have been proposed which have been effectively used for the task of drivable area segmentation as well.

It all started when Long et al. proposed fully convolutional networks for semantic segmentation [23]. They modified existing classification CNN architectures into FCN for semantic segmentation by fine-tuning. The major idea was that FCN can be taught mapping from pixels to pixels without the requirement of extracting region plans.

Since then, many ways have been proposed to improve the efficiency. In UNet, Olaf et al. proposed the conventional encoder and decoder design along with the use of strong data augmentation. In PspNet, Hengshuang et al. proposed the use of pyramid pooling module that provided a enhanced structure for pixel-level segmentation tasks. Recently, Daiqing et al. proposed the use of GANs along with semi-supervised learning for the task of semantic segmentation and achieved higher accuracy, in SemanticGAN [24]. Other techniques such as image analysis [25] and data augmentation [26] have also been explored for improving the efficiency the segmentation of drivable area.

C. Lane Line Segmentation Models

Due to the complexities associated with lane lines, such as the slimness of lane, the curvy nature of lanes, along with the disjointed pixel dispersal, the task of lane line segmentation requires much more detailed and complex architectures with precise detection skills.

Traditionally, Hough transform [27] was the most popular architecture used for lane line detection, but recent advancements in deep learning have led to the rise of many new networks for the task.

In LaneNet, Wang et al. . proposed a two staged deep neural network which used pixel-wise classification followed by lane line detection. In SCNN, a new concept of slice by slice convolutions rather than the contemporary deep layer by layer convolutions was proposed which allowed the information to pass through layers across pixels. Though this improved the accuracy, it was not time efficient. In ENet-SAD, a light weight architecture was proposed which utilized self-attention refining method which allowed the model to learn from itself, specifically from high level features to low level features as a form of guidance, which helped the model achieve high accuracy.

D. Multitasking Learning Models

Due to the need for a network with less computational needs and improved latency, many researchers have tried to develop a unified network to perform these three tasks. These networks also learn better due to the sharing of information among

different tasks. In Mask R-CNN [28], the network of Faster R-CNN is improved by inheriting RPN, utilizing the ResNet architecture and combining a third output arm for the efficient carrying out of object detection and segmentation tasks. In LNet, they proposed a three in one architecture with a new loss function, cross-IoU, aimed to suit object in distinct characteristics and dimensions. MultiNet came up the idea of using a pooled encoder and three different decoders for rectifying the tasks of object detection, scene classification and drivable area segmentation. Based on this, DLT-Net additionally added tensors to share information among different decoder heads. Though, Kang et al.[29], proved that this sharing of information is valuable only when the separate tasks of the decoders are related. Then, YOLOP also utilized the encoder decoder structure with 3 decoder heads to become the most effective multi-tasking system for these three tasks, and then based on this HybridNets proposed the use of BiFPN and the merging of drivable area segmentation head with the lane line segmentation head due to similarities between the both to improve the accuracy. Then, recently YOLOPv2 was released which improved the accuracy by utilizing a more fine-tuned bag of freebies along with a novel hybrid loss.

Though these multi-tasking models achieve good performance, none of them take into account the unstructured environment of the roads in rural areas and several parts of developing or underdeveloped countries, which makes the base for our model which takes into account these non-avoidable conditions.

III. METHODOLOGY

In this section, we discuss our approach for designing the feed forward network for accomplishing the tasks of drivable area segmentation, traffic object detection and lane segmentation, along with the optimizations for unstructured environments of the road. We propose a more effective architecture based on the works of YOLOPv2, HybridNets and YOLOP etc. Our model is inspired by the work of YOLOPv2 and HybridNets.

We hold the core design of the end-to-end HybridNets architecture which consists of one sharing encoder along with two different decoders to resolve different tasks, but utilize a much more efficient backbone and segmentation head which also takes into account the unstructured environment of the roads. The architecture of our model is shown in Figure 1.

A. Encoder

Our encoder comprises of a backbone network and a neck network. The backbone is a vital part of the architecture which extracts feature from the images that allows the networks to perform different types of tasks based on the different features extracted. Generally, image classification architectures with great accuracy are used as backbones for the extraction of features. The recently released YOLOv7, which achieves excellent detection accuracy along with remarkable speed improvement utilizes E-ELAN and therefore we have decided to choose E-ELAN as the backbone of our network. Our backbone reduces the computational cost along with utilizing group convolution so as to acquire numerous crucial features. P_1 exemplifies a feature level with resolution of $1/2^l$ of the input

pictures and the backbone only generates five feature maps and P_6 and P_7 are obtained from the reduction in size of P_5 .

Then, the neck network takes the feature maps from the backbone network where the key difficulty faced is multi-scale feature representations. Recently, BiFPN showed great improvement over FPN for this task, because of the restriction of FPN that information is only received in a one-way direction. This limitation is overcome by using BiFPN where, for every node, the features are fused at a distinct resolution in both the bottom-up and top-down path. Therefore, we use a BiFPN module which is derived from EfficientDet [30] as the neck network for our architecture to fuse features, where weights are added to each feature to realize the significance of every level.

B. Decoder

As shown in Figure 1, we designed two separate decoder heads, namely detection head and segmentation head. The detection head takes care of detecting the traffic objects such as cars and other vehicles traversing on the roads whereas the segmentation head takes of the drivable area segmentation and the lane line segmentation. We only use one head for the task of segmentation as the features of drivable area and the lane line are similar and generally the lane line serves as the border for the drivable area and thus it would be beneficial to use the same decoder for these tasks, as the sharing of information for these two tasks is maintained within the same decoder head. Because of the effectiveness of YOLOv7, we utilize an anchor centred multi-object detection structure. Path Aggregation Network (PAN) is used for better feature extraction because it follows a bottom-up structure. Then we use the semantic features from FPN which runs top-down and fuse these features together and then run detection in PAN for multiple fused features. Nine prior anchors are assigned to every grid of the multiple fused feature maps with diverse aspect ratios. Additionally, to create bounding boxes that cover all regions of various sizes, we choose 9 clusters and 3 different scales. Then, the detection head predicts the bounding boxes, along with its confidence value and the probability for each class.

Our segmentation head takes cares of drivable area segmentation as well as lane line segmentation. The output of our segmentation head are three classes, namely lane line, background and drivable area. The features of deep layers are provided to the segmentation head for better feature fusion that helps improve the precision of the network. The features P_3 to P_7 from the neck are upsampled to have same size of the output feature map. Then these features are combined with features from P_2 passed through a convolutional block to have same size feature maps, by summing all levels for better fusion of features. Finally, we resize the output feature to $(W, H, 3)$, which represents the probability of each belonging pixel class. Our segmentation head also takes into account the features of unstructured environments of roads such as potholes, muddy areas, etc. to provide better segmentation for unstructured environments. The architecture of our segmentation head is shown in Figure 2.

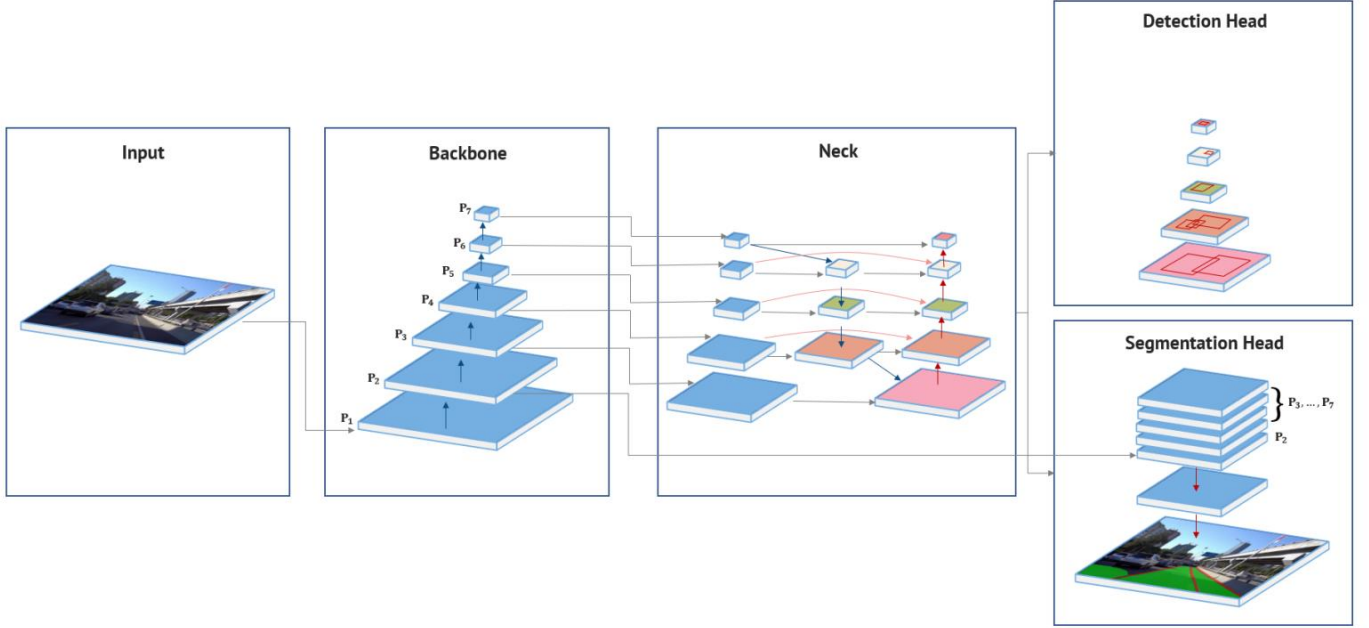


Figure 1. DrivableNets Architecture has one encoder consisting of a backbone and neck network, and two decoders consisting of Segmentation and Detection Head.

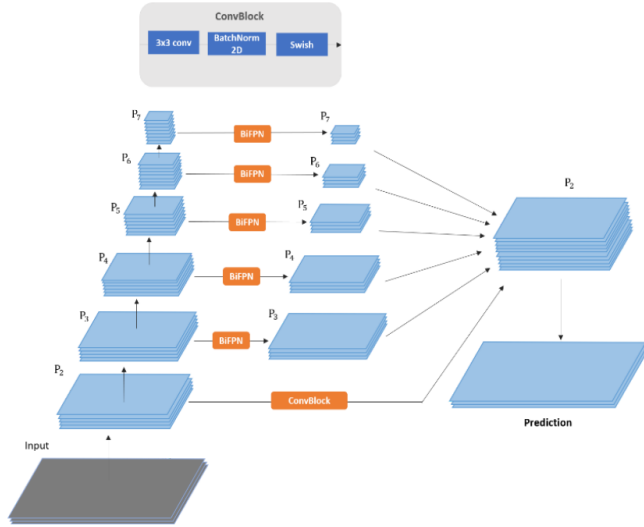


Figure 2. Segmentation Head for DrivableNets.

IV. EXPERIMENTATION AND EVALUATION

This section discusses the dataset characteristics and the various parameter configurations used in our experiments along with the evaluation of our model. All experiments were carried out on a PC with graphics card NVIDIA RTX A5000 and torch 1.12.

We used the Indian Driving Dataset for the training and validation of the model. The dataset contains images and videos of roads in unstructured environments and structured environments. We use 10K images for training, 2K for

validation and 3K images for testing. We reorganise our dataset in the format of the popular BDD100K dataset so as to help in comparison and for the training of the architecture. The object detection model is trained for detecting car, bus, and trucks and they are all merged into a single class labelled ‘vehicle’. Drivable class is formed after merging direct and alternative segmentation class. For annotating the lane lines, we use a pixel width of 4 to converge two lane line annotations into one. The size of the input images are 640 X 384, so as to make sure there is a good relation between the accuracy and the performance of the model and to maintain that the dimension of image are divisible by 128 as per the requirements of BiFPN. Data Augmentation techniques are also used to improve the accuracy of our model, such as scaling, flipping, rotating and mosaic augmentation is used while training detection decoder. Adam optimizer is used for training with learning rate of $1e^{-3}$. The model utilizes an IoU limit of 0.5 for big annotations and 0.3 for smaller annotations. We train our model for 200 epochs with a batch size of 32.

A. Traffic Object Detection Evaluation

For the task of traffic object detection, recall and mAP50 are utilized as evaluation metrics. Our model achieves higher mAP and recall than the latest SOTA multi-tasking models, as shown in Table 1.

Network	Recall	mAP50
MultiNet	81.3	60.2
DLT-Net	89.4	68.4
HybridNets	92.8	77.3
YOLOP	89.2	76.5
YOLOPv2	91.1	83.4
DrivableNets	93.1	85.3

Table 1. Comparison of traffic object detection results.

B. Lane Line Segmentation Evaluation

For the task of lane line detection, we use IoU of lane lines and pixel accuracy as the evaluation metrics. Our model achieved the highest accuracy, as shown in Table 2.

Network	Accuracy	Lane Line IoU
ENet	34.12	14.64
SCNN	35.79	15.84
ENet-SAD	36.56	16.02
HybridNets	85.40	31.60
YOLOP	70.50	26.20
YOLOPv2	87.31	27.25
DrivableNets	88.41	28.32

Table 2. Comparison of lane line segmentation results.

C. Drivable Area Segmentation Evaluation

The results of the drivable area segmentation are shown in Table 3. The metric used to evaluate the performance of our segmentation model is mIoU. Our model just slightly pales in comparison to that of YOLOPv2 as they employed three different decoders while we only use two.

Network	mIoU
MultiNet	71.6
DLT-Net	71.3
PSPNet	89.6
HybridNets	90.5
YOLOP	91.5
YOLOPv2	93.2
DrivableNets	92.8

Table 1. Comparison of drivable area segmentation results.

Some of the results of our model are visualised in the following images, on both structured and unstructured environments, for all the three tasks of traffic object detection, lane line segmentation and drivable area segmentation.



Figure 3. Visualisation of our model in unstructured environment.



Figure 4. Visualisation of our model in unstructured environment.



Figure 5. Visualisation of our model in unstructured environment.



Figure 6. Visualisation of our model in structured environment.

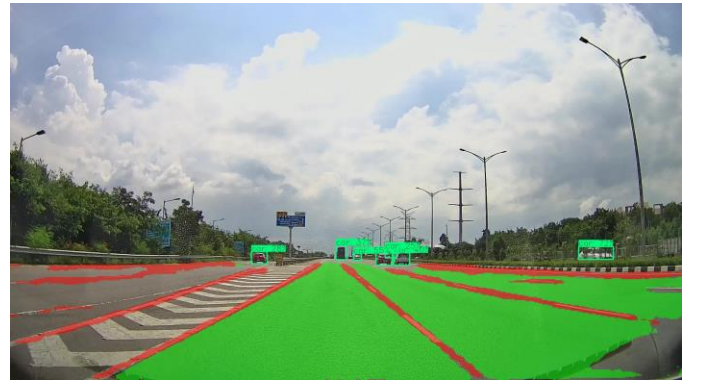


Figure 7. Visualisation of our model in structured environment.

Few things can be noted from these visualisations of our model's results. Our model performs well in both structured and unstructured environments, our model is able to segment out

lane lines of various types, our model is able to segment a drivable area even when unexpected obstacles such as humans are present on the road jay walking and our model is capable of excluding the muddy areas, or wet potholes into our drivable area as shown in Figure 3.

V. CONCLUSION AND FUTURE WORK

In this paper, we reviewed existing multi-tasking approaches for detecting the drivable area and found out the limitations of all the existing models and proposed a new model, DrivableNets, that takes into account the unstructured environments of road and achieves higher or comparable efficiency relative to the current models.

In future works, we would like to refine our model for the unstructured environments and be able to classify a non-drivable area also, which will aid in autonomous driving and several tasks such as estimating the parts of the roads that needs to be repaired etc. We would also like to ameliorate the context of structures in lane lines and drivable area and also improve the accuracy further.

REFERENCES

- [1] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks". *Advances in neural information processing systems*, 28, 2015.
- [2] W. Liu et al., "SSD: Single Shot MultiBox Detector". In: Leibe, B. et al. (eds.) *Computer Vision – ECCV 2016*. pp. 21-37 Springer International Publishing, Cham, 2016, https://doi.org/10.1007/978-3-318-46338-0_2.
- [3] Kaiwen Duan, Song Bai, Lingxi Xie, Honggang Qi, Qingming Huang, and Qi Tian. Centernet: Keypoint triplets for object detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2019.
- [4] Redmon J, Divvala S, Girshick R, Farhadi A, "You only look once: Unified, real-time object detection". In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2016. Pp. 779-788.
- [5] Joseph Redmon and Ali Farhadi, "Yolov3: An incremental improvement". *arXiv preprint arXiv:1804.02767*, 2018.
- [6] Alexey Bochkovskiy, Chien-Yao Wang, Hong-Yuan Mark Liao, "Yolov4: Optimal Speed and Accuracy of Object Detection". *arXiv:2004.10934v1*.
- [7] Chien-Yao Wang, Alexey Bochkovskiy, and HongYuan Mark Liao, "Yolov7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors". *arXiv preprint arXiv:2207.02696*, 2022.
- [8] Huimin Huang, Lanfen Lin, Ruofeng Tong, Hongjie Hu, Qiaowei Zhang, Yutaro Iwamoto, Xianhua Han, Yen-Wei Chen, and Jian Wu, "Unet 3+: A full-scale connected unet for medical image segmentation". In *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1055–1059. IEEE, 2020.
- [9] Vijay Badrinarayanan, Alex Kendall, and Roberto Cipolla. Segnet, "A deep convolutional encoder-decoder architecture for image segmentation". *IEEE transactions on pattern analysis and machine intelligence*, 39(12):2481–2495, 2017.
- [10] Hengshuang Zhao, Jianping Shi, Xiaojuan Qi, Xiaogang Wang, and Jiaya Jia, "Pyramid scene parsing network". In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2881–2890, 2017.
- [11] Z. Wang, W. Ren, Q. Qiu, "LaneNet: Real-Time Lane Detection Networks for Autonomous Driving." *arXiv:1807.01726*, 2018.
- [12] X. Pan, J. Shi, P. Luo, X. Wang, X. Tang, "Spatial As Deep: Spatial CNN for Traffic Scene Understanding." *arXiv:1712.06080*, 2017.
- [13] Y. N. Hou, Z. Ma, C. X. Liu, C. C. Loy, "Learning lightweight lane detection CNNs by self attention distillation". In *Proceedings of IEEE/CVF International Conference on Computer Vision, IEEE, Seoul, Korea*, pp.1013– 1021, 2019. DOI: 10.1109/ICCV.2019.00110.
- [14] Y. Qian, J. M. Dolan and M. Yang, "DLT-Net: Joint Detection of Drivable Areas, Lane Lines, and Traffic Objects," *IEEE Transactions on Intelligent Transportation Systems*, vol. 21, no. 11, pp. 4670-4679, Nov. 2020, doi: 10.1109/TITS.2019.2943777.
- [15] M. Teichmann, M. Weber, M. Zöllner, R. Cipolla and R. Urtasun, "MultiNet: Real-time Joint Semantic Reasoning for Autonomous Driving," in *Proceedings of the IEEE Intelligent Vehicles Symposium (IV)*, 2018, pp. 1013-1020, doi: 10.1109/IVS.2018.8500504.
- [16] Dat Vu, Bao Ngo, and Hung Phan, "Hybridnets: End-to-end perception network". *arXiv preprint arXiv:2203.09035*, 2022.
- [17] Dong Wu, Manwen Liao, Weitian Zhang, and Xinggang Wang, "Yolop: You only look once for panoptic driving perception". *arXiv preprint arXiv:2108.11250*, 2021.
- [18] Cheng Han, Qichao Zhao, Shuyi Zhang, Yinzi Chen, Zhenlin Zhang and Jinwei Yuan, "YOLOPv2: Better, Faster, Stronger for Panoptic Driving Perception". *arXiv preprint arXiv:2208.11434*, 2022.
- [19] G. Varma, A. Subramanian, A. Namboodiri, M. Chandraker and C. V. Jawahar, "IDD: A Dataset for Exploring Problems of Autonomous Navigation in Unconstrained Environments," *2019 IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2019, pp. 1743-1751, doi: 10.1109/WACV.2019.00190.
- [20] F. Yu, W. Q. Xian, Y. Y. Chen, F. C. Liu, M. K. Liao, V. Madhavan, T. Darrell, "BDD100K: A diverse driving video database with scalable annotation tooling". DOI: 10.48550/ arXiv.1805.04687, 2018.
- [21] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan and S. Belongie, "Feature Pyramid Networks for Object Detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 936-944, doi: 10.1109/CVPR.2017.106.
- [22] Jifeng Dai, Yi Li, Kaiming He, Jian sun, "R-FCN: Object Detection via Region-based Fully Convolutional Networks". *arXiv preprint arXiv:1605.06409*, 2016.
- [23] J. Long, E. Shelhamer, T. Darrell, "Fully convolutional networks for semantic segmentation". In *Proceedings of IEEE Conference on Computer Vision and Pattern Recognition, IEEE, Boston, USA*, pp.3431– 3440, 2015. DOI: 10.1109/ CVPR.2015.7298965.
- [24] D. Li, J. Yang, K. Kreis, A. Torralba and S. Fidler, "Semantic Segmentation with Generative Models: Semi-Supervised Learning and Strong Out-of-Domain Generalization," *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 8296-8307, doi: 10.1109/CVPR46437.2021.00820.
- [25] K.I. Kiy, "A New Method of Global Image Analysis and Its Application in Understanding Road Scenes," *Pattern Recognit. Image Anal.* 28, 3, 483–495, 2018, <https://doi.org/10.1134/S1054661818030100>.
- [26] J. Muñoz-Bulnes, C. Fernandez, I. Parra, D. Fernández-Llorca and M. A. Sotelo, "Deep fully convolutional networks with random data augmentation for enhanced generalization in road detection," in *Proceedings of the IEEE 20th International Conference on Intelligent Transportation Systems (ITSC)*, 2017, pp. 366-371, doi: 10.1109/ITSC.2017.8317901.
- [27] F. Zheng, S. Luo, K. Song, C-W. Yan, M-C. Wang, "Improved Lane Line Detection Algorithm Based on Hough Transform," *Pattern Recognit. Image Anal.* 28, 254–260, 2018.
- [28] K. He, G. Gkioxari, P. Dollár and R. Girshick, "Mask R-CNN," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 42, no. 2, pp. 386-397, 1 Feb. 2020, doi: 10.1109/TPAMI.2018.2844175.
- [29] Z. L. Kang, K. Grauman, F. Sha, "Learning with whom to share in multi-task feature learning". In *Proceedings of the 28th International Conference on Machine Learning*, Bellevue, USA, pp.521–528, 2011.
- [30] M. Tan, R. Pang and Q. V. Le, "EfficientDet: Scalable and Efficient Object Detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 10778-10787, doi: 10.1109/CVPR42600.2020.01079.